

Exercise I.

[20 marks]

Compute the following integrals:

1. $\int \frac{\sin^3(\ln|x|) \cos^3(\ln|x|)}{x} dx,$

3. $\int x^2 \arctan x dx,$

2. $\int \frac{x^3}{x^2 + 1} dx,$

4. $\int \frac{4x + 1}{\sqrt{x^2 + x + 1}} dx.$

Exercise II.

[20 marks]

Study the natures of the following series:

1. $\sum_{n=2}^{\infty} \frac{1}{n \ln n},$

3. $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n} \arctan n,$

2. $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n},$

4. $\sum_{n=2}^{\infty} \frac{1}{n \ln n} \arctan n.$

Exercise III.

[20 marks]

Let $\alpha \in \mathbb{R}$. Study the natures of the following improper integrals:

1. $\int_1^{\infty} \frac{1}{1+x^\alpha} dx,$

3. $\int_e^{\infty} \frac{\sin x}{\ln x} dx,$

2. $\int_0^1 \frac{1}{(\sin x)^\alpha \sqrt{1-x}} dx,$

4. $\int_0^{\frac{\pi}{2}} \sqrt{\tan x} dx.$

Exercise IV.

[25 marks]

1. Calculate the following limit:

$$\lim_{n \rightarrow \infty} \sqrt[n]{\prod_{k=1}^n \frac{n^2 + k^2}{n^2}}.$$

2. Let $g(x) = \int_0^{\sin x} e^{t^2} dt$. Compute $g'(x)$.

3. Calculate the following limit:

$$\lim_{x \rightarrow \pi} \frac{1}{x - \pi} \int_0^{\sin x} e^{t^2} dt.$$

Exercise V.

[15 marks]

Let $I_n = \int_{-\infty}^{\infty} x^{2n} e^{-\frac{x^2}{2}} dx$. Given that $I_0 = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx = \sqrt{2\pi}$.

1. Show that the improper integral I_n is convergent for all $n \in \mathbb{N}$.

2. Show that $I_{n-1} = \frac{1}{(2n-1)} I_n$ for all $n \in \mathbb{N}^*$ and deduce the value of I_n .

Exercise I. Compute the following integrals:

$$\begin{aligned}
 1. \int \frac{\sin^3(\ln|x|) \cos^3(\ln|x|)}{x} dx &\stackrel{(1)}{=} \int \sin^3 t \cos^3 t dt ; \ln|x| = t \Rightarrow \frac{1}{x} dx = dt \\
 &\stackrel{(1)}{=} \int \sin^3 t \cos^2 t \cos t dt ; \sin t = u \Rightarrow \cos t dt = du \\
 &\stackrel{(1)}{=} \int u^3 (1-u^2) du \\
 &= \frac{u^4}{4} - \frac{u^6}{6} + C = \frac{\sin^4 t}{4} - \frac{\sin^6 t}{6} + C \\
 &\stackrel{(1)}{=} \frac{\sin^4(\ln|x|)}{4} - \frac{\sin^6(\ln|x|)}{6} + C
 \end{aligned}$$

$$\begin{aligned}
 2. \int \frac{x^3}{x^2+1} dx &\stackrel{(1)}{=} \int \left[x - \frac{x}{x^2+1} \right] dx = \int x dx - \int \frac{x}{x^2+1} dx ; \frac{x^3}{x^2+1} = \frac{x^3}{x^2+1} \cdot \frac{x^2+1}{x^2+1} \stackrel{(2)}{=} \frac{x^3(x^2+1)}{x^2+1} \\
 &\stackrel{(1)}{=} \frac{x^2}{2} - \frac{1}{2} \int \frac{d(x^2+1)}{x^2+1} \stackrel{(1)}{=} \frac{x^2}{2} - \frac{1}{2} \ln|x^2+1| + C
 \end{aligned}$$

$$\begin{aligned}
 3. \int \frac{x^2 \arctan x}{\sqrt{1+x^2}} dx &\stackrel{(4)}{=} \frac{x^3}{3} \arctan x - \frac{1}{3} \int \frac{x^3}{1+x^2} dx ; u = \arctan x \quad \begin{array}{l} v' = x^2 \\ u' = \frac{1}{1+x^2} \end{array} \quad \begin{array}{l} x \\ -1 \end{array} \quad \begin{array}{l} v = \frac{x^3}{3} \\ u = \frac{1}{1+x^2} \end{array} \\
 &\stackrel{(1)}{=} \frac{x^3}{3} \arctan x - \frac{1}{6} x^2 + \frac{1}{6} \ln|x^2+1| + C \quad (\text{using 2.})
 \end{aligned}$$

$$\begin{aligned}
 4. \int \frac{4x+1}{\sqrt{x^2+x+1}} dx &\stackrel{(1)}{=} 2 \int \frac{2x+\frac{1}{2}+\frac{1}{2}-\frac{1}{2}}{\sqrt{x^2+x+1}} dx \stackrel{(1)}{=} 2 \int \frac{2x+\frac{1}{2}}{\sqrt{x^2+x+1}} dx - \int \frac{dx}{\sqrt{x^2+x+1}} \\
 &\stackrel{(1)}{=} 4 \sqrt{x^2+x+1} - \int \frac{d(x+\frac{1}{2})}{\sqrt{(x+\frac{1}{2})^2+\frac{3}{4}}} \stackrel{(1)}{=} 4 \sqrt{x^2+x+1} - \ln \left(\left[x+\frac{1}{2} \right] + \sqrt{(x+\frac{1}{2})^2+\frac{3}{4}} \right) + C
 \end{aligned}$$

Exercise II. Study the natures of the following series:

$$1. \sum_{n=2}^{\infty} \frac{1}{n \ln n} \quad \text{div, Bertrand example } \alpha=1 \text{ \& } p=1 \leq 1.$$

$$2. \sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n} \quad \text{Conv, Alternating Series } \frac{1}{n \ln n} \text{ dec and } \frac{1}{n \ln n} \xrightarrow{n \rightarrow \infty} 0.$$

$$3. \sum_{n=2}^{\infty} \frac{(-1)^n \arctan n}{n \ln n} \quad \text{Conv, Abel's test } \left\{ \begin{array}{l} \sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n} \text{ conv} \\ b_n = \arctan n \text{ is } \uparrow \text{ and } \text{bdd " } |b_n| \leq \frac{\pi}{2} \text{ " } \end{array} \right.$$

$$4. \sum_{n=2}^{\infty} \frac{1}{n \ln n} \arctan n \quad \text{div, comparison test } \frac{1}{n \ln n} \arctan n \geq \arctan 2 \cdot \frac{1}{n \ln n} > 0$$

Exercise III. let $\alpha \in \mathbb{R}$. study the natures of the following improper integrals:

$$1. \int_1^{\infty} \frac{1}{1+x^\alpha} dx \quad o < \frac{1}{1+x^\alpha} \underset{\infty}{\sim} \begin{cases} \frac{1}{x^\alpha} & \text{if } \alpha > 0 \\ \frac{1}{2} & \text{if } \alpha = 0 \\ 1 & \text{if } \alpha < 0 \end{cases}$$

The imp int conv^③ for $\alpha > 1$ (Riemann), and div^② for $\alpha \leq 1$.

$$2. \int_0^1 \frac{1}{(\sin x)^\alpha \sqrt{1-x}} dx = \int_0^{1/2} + \int_{1/2}^1 = I_1 + I_2 \quad ①$$

$$\text{near } 0^+: o < \frac{1}{(\sin x)^\alpha \sqrt{1-x}} \underset{0^+}{\sim} \frac{1}{x^\alpha}; \quad I_1 \text{ conv iff } \alpha < 1 \quad ②$$

$$\text{near } 1^-: o < \frac{1}{(\sin x)^\alpha \sqrt{1-x}} \underset{1^-}{\sim} \frac{1}{(\sin 1)^\alpha} \cdot \frac{1}{(1-x)^{1/2}}; \quad I_2 \text{ conv (Riem)} \quad ②$$

Thus I conv iff $\alpha < 1$.

$$3. \int_e^{\infty} \sin x \cdot \frac{1}{\ln x} dx \quad \text{conv, Abel-Dirichlet} \quad ① \quad \begin{cases} \bullet \frac{1}{\ln x} \underset{x \rightarrow \infty}{\xrightarrow{0}} \text{ (monotone)} \quad \frac{1}{\ln x} \underset{x \rightarrow \infty}{\xrightarrow{0}} \\ \bullet |F(x)| = \left| \int_e^x \sin x dx \right| \leq 2, F \text{ is bdd on } [e, \infty[\end{cases} \quad ②$$

$$4. \int_0^{\pi/2} \sqrt{\tan x} dx \stackrel{③}{=} \int_0^{\infty} \frac{\sqrt{t}}{1+t^2} dt$$

$$\bullet \frac{\sqrt{t}}{1+t^2} \underset{t \rightarrow \infty}{\sim} \frac{t^{1/2}}{t^2} = \frac{1}{t^{3/2}} > 0 \quad ①$$

$$\begin{aligned} & x = u(t) = \arctan t, \quad u' = \frac{1}{1+t^2} \\ & t = \tan x = f(u) \in C([0, \frac{\pi}{2}[) \end{aligned}$$

By equiv let $\int_0^{\infty} \frac{\sqrt{t}}{1+t^2} dt$ is conv^①. Thus $\int_0^{\pi/2} \sqrt{\tan x} dx$ is conv.

Exercise IV.

1. Calculate the following limit:

$$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n \frac{n^2+k^2}{n^2} \right)^{1/n} \stackrel{②}{=} \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln \left(\prod_{k=1}^n \frac{n^2+k^2}{n^2} \right)} \stackrel{②}{=} \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln \left(1 + \left(\frac{k}{n} \right)^2 \right)}$$

$$\stackrel{①}{=} e^{\int_0^1 \frac{\ln(1+x^2)}{x} dx}$$

"f.e.c.([0,1]), Hence
F.e.R.([0,1])"

$$\stackrel{②}{=} e^{\left[x \ln(1+x^2) \right]_0^1 - 2 \int_0^1 \frac{x^2}{1+x^2} dx}$$

$$\stackrel{①}{=} \ln 2 - 2 \left[1 - \arctan x \right]_0^1$$

$$\stackrel{①}{=} e$$

$$\stackrel{①}{=} e^{\ln 2 - 2 + \frac{\pi}{4}}$$

2. let $g(x) = \int_0^{\sin x} e^{t^2} dt$. Compute $g'(x)$.

$$g'(x) = (\sin x)' e^{(\sin x)^2} = \cos x e^{\sin^2 x}. \quad ⑤$$

3. calculate $\lim_{x \rightarrow \pi} \frac{1}{x-\pi} \int_0^{\sin x} e^{t^2} dt$.

$$\lim_{x \rightarrow \pi} \frac{1}{x-\pi} \int_0^{\sin x} e^{t^2} dt \stackrel{(3)}{=} \lim_{x \rightarrow \pi} \frac{1}{x-\pi} e^{\int_0^{\sin x} 1 dx}$$

$$\stackrel{(1)}{=} \lim_{x \rightarrow \pi} \frac{\sin x}{x-\pi} = \lim_{x \rightarrow \pi} \frac{\sin(\pi-x)}{-(\pi-x)} \stackrel{(2)}{=} -1$$

• $f(t) = e^{t^2}$ is cont on $[0, \sin x]$

• $g(t) = 1 > 0$ on $[x, 2x]$

• $0 < c < \sin x$ as $x \rightarrow \pi$, $c \rightarrow 0$

or one can use 2. $\lim_{x \rightarrow \pi} \frac{g(x)}{x-\pi} \stackrel{(5)}{=} \lim_{x \rightarrow \pi} \frac{g(x) - g(\pi)}{x-\pi} \stackrel{(2)}{=} g'(\pi) \stackrel{(2)}{=} \cos \pi e^{\sin^2 \pi} \stackrel{(1)}{=} -1$.

or one can use Hopital Rule!

Exercise V. let $I_n = \int_{-\infty}^{\infty} x^{2n} e^{-\frac{x^2}{2}} dx$. Given $I_0 = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx = \sqrt{2\pi}$.

1. Show that I_n is convergent $\forall n \in \mathbb{N}$.

$$x^{10} \cdot \frac{x^{2n}}{e^{\frac{x^2}{2}}} \xrightarrow{x \rightarrow \pm \infty} 0 \text{ and } x = 10 > 1 \stackrel{(5)}{\Rightarrow} I_n \text{ conv } \forall n \in \mathbb{N}.$$

2. Show that $I_{n-1} = \frac{1}{(2n-1)} I_n$ for all $n \in \mathbb{N}^*$ and deduce the value of I_n .

$$\begin{aligned} \bullet I_{n-1} &= \int_{-\infty}^{\infty} \underbrace{x^{2n-2}}_{u'} \underbrace{e^{-\frac{x^2}{2}}}_{v} dx \stackrel{(2)}{=} \frac{x^{2n-1}}{2n-1} e^{-\frac{x^2}{2}} \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{x^{2n-1}}{2n-1} \cdot (-x) e^{-\frac{x^2}{2}} dx \\ &= \frac{1}{2n-1} \int_{-\infty}^{+\infty} x^{2n} e^{-\frac{x^2}{2}} dx \stackrel{(1)}{=} \frac{1}{2n-1} I_n. \end{aligned}$$

$$\bullet I_n \stackrel{(1)}{=} (2n-1) I_{n-1} \stackrel{(1)}{=} (2n-1)(2n-3) I_{n-2} = (2n-1)(2n-3)(2n-5) \dots 3 \cdot 1 \cdot I_0$$

$$\stackrel{(3)}{=} 1 \cdot 3 \cdot 5 \dots (2n-5)(2n-3)(2n-1) \cdot \sqrt{2\pi}.$$